

Non-Discriminatory Personalized Pricing

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Motivation

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How should a seller set prices given that some consumer characteristics cannot be discriminated?

What are the welfare implications?

FINANCE

Regulators caught Wells Fargo, other banks in probe over mortgage pricing discrimination

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Hugh Son
@HUGH_SON

KEY POINTS

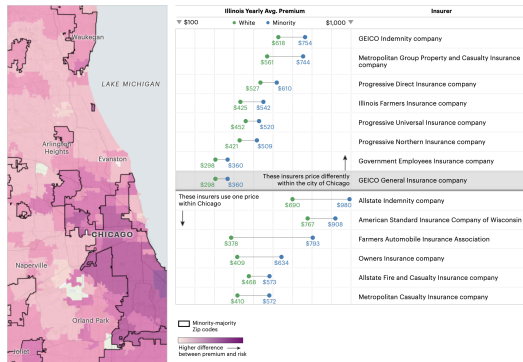
- Wells Fargo received an official notice from the Consumer Financial Protection Bureau on problems with its use of mortgage rate discounts, sources said.
- Wells Fargo hired a law firm to grill mortgage bankers whose sales included high levels of the discounts, said the sources.
- Several banks received MRAs about lending practices last year, the CFPB said without naming any of the institutions.
- In their industry review, regulators found "statistically significant disparities" in the rates in which Black and female borrowers got pricing exceptions compared with other customers.

Chicago Area Disparities in Car Insurance Premiums

By Al Shaw, Jeff Larson and Julia Angwin, ProPublica, April 5, 2017

Some car insurers charge higher premiums in Chicago's minority neighborhoods than in predominantly white neighborhoods with similar insurance losses. The areas outlined in black are more than 50 percent minority. Many insurers charge the same premiums throughout Chicago, but quote higher prices than in suburbs with similar risk. | [Read the methodology](#) » | [Related story: Minority Neighborhoods Pay Higher Car Insurance Premiums Than White Areas With the Same Risk](#) »

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MACHINE BIAS



The Tiger Mom Tax: Asians Are Nearly Twice as Likely to Get a Higher Price from Princeton Review

by Julia Angwin, Surya Mattu and Jeff Larson, Sept. 1, 2015, 10 a.m. EDT

Protected-characteristic cases in personalized pricing

- **Tinder (age):** older users were quoted higher prices than younger users.
- **U.S. auto dealers (race / ethnicity):** cases alleged Black, Hispanic, and Latino buyers paid more for the same add-ons.
- **Princeton Review (ZIP code proxy):** SAT-prep quotes varied by ZIP code; Asian households were more likely to get high-price quotes.
- **UK motor insurance (ethnicity):** people of colour were found to pay more on average, suggesting indirect discrimination.

Takeaway: Direct protected-trait cases are relatively rare; many pricing differences appear through proxies such as ZIP code or geography.

Fraction of U.S. Transactions Under Non-Discrimination Constraints

- **Range:** $\approx 36\% - 66\%$ of consumer transaction value
- **Lower bound (36%):** Health (Affordable Care Act (ACA) / Americans with Disabilities Act (ADA)), Food & accommodation (Public Accommodations / ADA), Recreation (Public Accommodations / ADA), Housing rent (Fair Housing Act)
- **Upper bound (60–66%):** Retail & food services + Health + Housing rent (broad ADA / public-facing business coverage; excluding e-commerce $\Rightarrow \sim 60\%$)
- **Data:** U.S. Personal Consumption Expenditures (Bureau of Economic Analysis (BEA); Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis, 2025)

Market Segmentations and (Third Degree) Price Discrimination: Varian (1985); Aguirre, Cowan and Vickers (2010); Cowan (2016); Bergemann, Brooks and Morris (2015) Kallus and Zhou (2021); Cohen, Elmachoub and Lei (2022); Haghpanah and Siegel (2022, 2023); Farboodi, Haghpanah and Shourideh (2025).

Privacy Preserving Signals: He, Sandomirskiy and Tamuz (2024); Strack and Yang (2024).

Optimal Transport in Mechanism/Information Design: Doval and Eliat (2023); Arieli, Babichenko and Sandomirskiy (2023); Skreta and Perez-Richet (2024); Kolotilin, Corrao and Wolitzky (2024); Ball and Kattwinkel (2024); Lin and Liu (2024); Echenique, Root and Sandomirskiy (2024); Moscariello (2025);

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Assumption:

Seller is restricted to pricing rules that do not “discriminate” based on the protected characteristic θ .

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- In the US companies legally required to collect protected characteristics.

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*“Big data naturally raises concerns among groups that have historically been victims of discrimination. Given hundreds of variables to choose from, it is easy to imagine that statistical models could be used to hide more explicit forms of discrimination by generating customer segments that are **closely correlated with race, gender, ethnicity, or religion** [...], even if the profit motive is different from [...] the sort of prejudice that our antidiscrimination laws seek to prohibit.”*

— The White House, 2015, Big Data and Differential Pricing.

Correlation as a Measure of Discrimination

In practice (in the US): checking discrimination based on whether prices are statistically correlated with characteristics.

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"[...] The [ECO] Act and regulation may prohibit a creditor practice that is discriminatory in effect because it has a disproportionately negative impact on a prohibited basis, even though the creditor has no intent to discriminate and the practice appears neutral on its face."

— 12 CFR Regulation B

Non-Discriminatory Pricing

A pricing rule p is **discriminatory** if for consumer with the same cost c the distribution of prices depends on the protected characteristic θ :

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Man/woman who are equally costly to serve should face the same (distribution of) prices.

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- Why condition on c ? seller is allowed to condition on materially relevant information (credit score, cost, ..). Being predictive is **not** enough to be materially relevant.
- All our results go through if there is additional materially relevant information.
- Discrimination need not be intentional.
- This notion is used in the US in the context of anti-discrimination law (e.g., ECOA, FCRA, CRA, FHA, CA-AB1287) and in several OECD countries.
- Alternative notions later in the talk.

Aside: Parctical Enforcement

- From Consumer Financial Protection Bureau (CFPB) Issue 25 (2021).
- “Examiners observed that mortgage lenders violated ECOA and Regulation B by discriminating against African American and female borrowers in the granting of pricing exceptions based upon competitive offers from other institutions.”
- “The examination team identified lenders with statistically significant disparities for the incidence of pricing exceptions for African American and female applications compared to similarly situated non-Hispanic white and male borrowers. Examiners did not identify evidence that explained the disparities observed in the statistical analysis.”
- Most violations are offering different interest rates to people with the same default risk/credit score!
- Aside: firms have to report all offers to CFPB and are required to collect protected characteristics.

Optimal Non-Discriminatory Pricing Rule

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Focus on Pareto undominated optimal pricing rules (= there does not exist another pricing rules that induce higher profit, higher average surplus among consumers with $\theta = l$ and $\theta = h$).

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 - Sell to the $\theta = l$ consumer only and get a profit of $(1 - \alpha_c)(v_l - c)^+$.
- Optimal profit $\pi_c(v_l, v_h) := \max\{\min\{v_l, v_h\} - c, \alpha_c(v_h - c)^+\}$.

Optimal Pricing as a Matching Problem

- Let $\alpha_c = \mathbb{P}[\theta = h \mid c]$. Within each pair (v_l, v_h) , the seller can
 - Sell to both and get a profit of $\min\{v_l, v_h\} - c$,
 - Sell to the $\theta = h$ consumer only and get a profit of $\alpha_c(v_h - c)^+$, or
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Optimal Pricing as Optimal Transport

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Proposition (Optimal Transport Representation)

Let π^* be the value of the optimal transport problem:

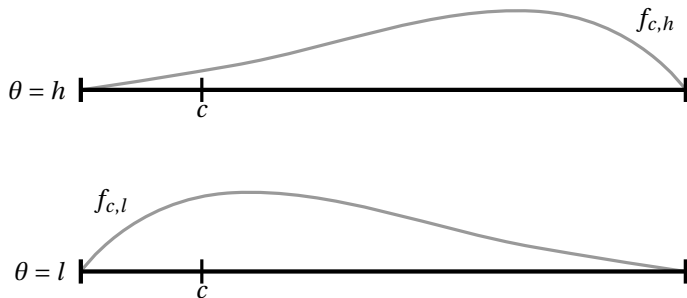
$$\pi^* := \int_C \left(\max_{\rho_c} \int_{V^2} \pi_c(v_l, v_h) d\rho_c(v_l, v_h) \right) dG(c). \quad (\text{T})$$

Then $\pi^* = \Pi^*$. Moreover, any solution of (P) induces a solution of (T), and any solution of (T) induces a solution of (P).

Example: Assortative Matching

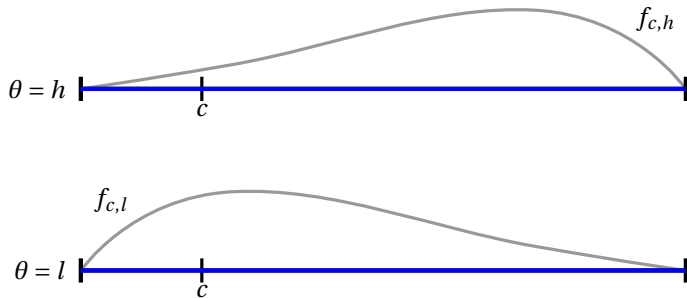
Non-Discriminatory Pricing as Matching

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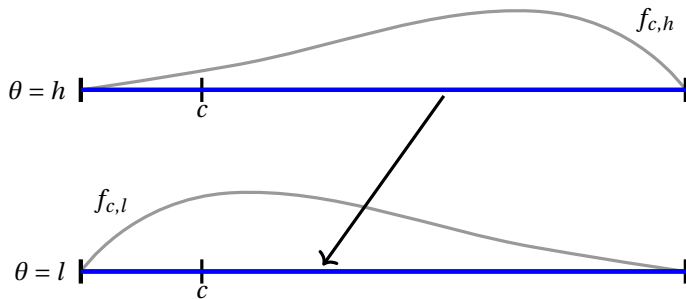
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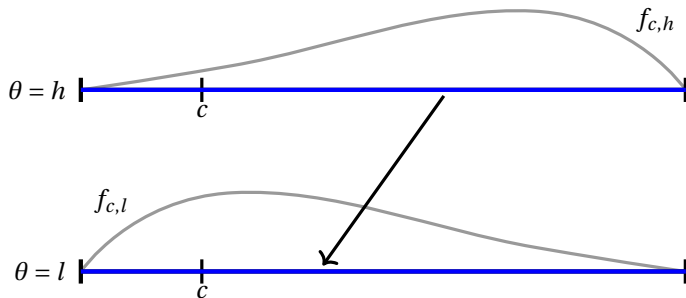
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$$p^{ass}(v, c, \theta, r) = \begin{cases} \max\{(F_{c,l}^{-1} \circ F_{c,h})(v), c\} & \text{if } \theta = h \\ \max\{v, c\} & \text{if } \theta = l \end{cases}$$

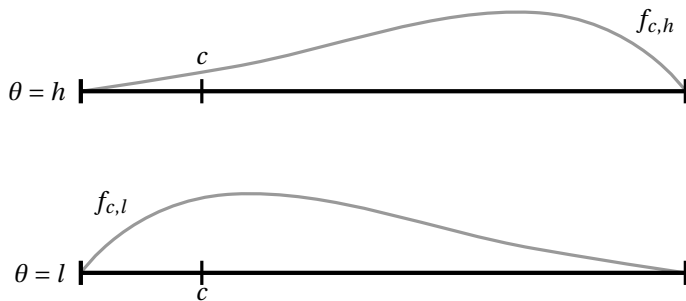
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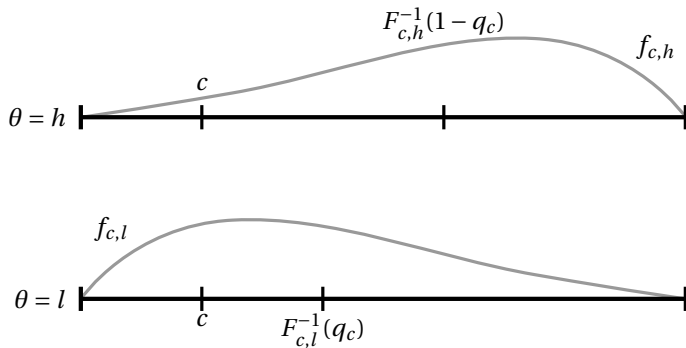
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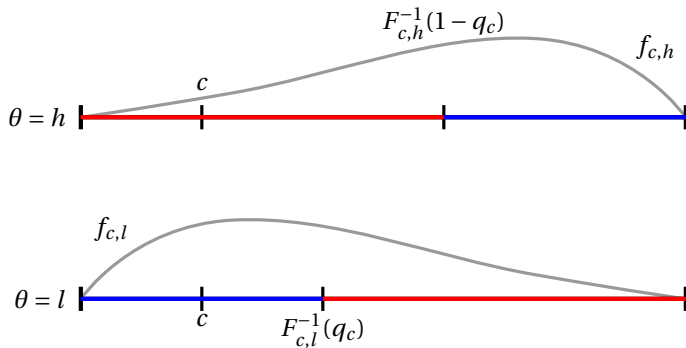
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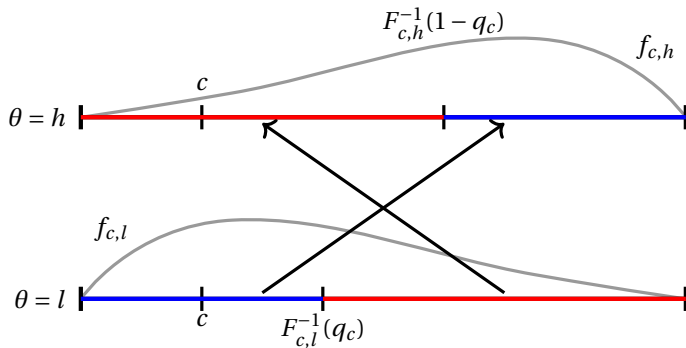
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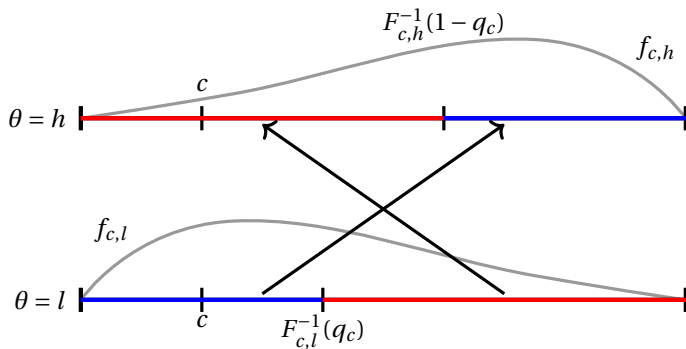
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$$p^{anti}(v, c, \theta, r) = \begin{cases} \max\{F_{c,h}^{-1}(F_{c,l}(v) - q_c), c\} & \text{if } \theta = l \text{ and } v > F_{c,l}^{-1}(q_c) \\ \max\{F_{c,h}^{-1}(F_{c,l}(v) + 1 - q_c), c\} & \text{if } \theta = l \text{ and } v \leq F_{c,l}^{-1}(q_c) \\ \max\{v, c\} & \text{if } \theta = h \end{cases}$$

Structure of the Optimal Transport

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Equivalent to minimizing available surplus – profit:

$$0.5(v_l + v_h) - \max\{0.5 \max\{v_l, v_h\}, \min\{v_l, v_h\}\} = 0.5 \min\{v_l, v_h, |v_h - v_l|\}.$$

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This objective is non-convex, non-concave, and not translation invariant.

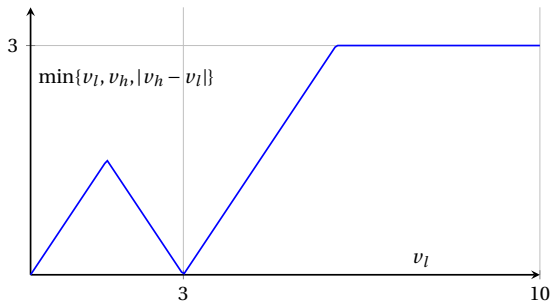
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$$0.5(v_l + v_h) - \max\{0.5 \max\{v_l, v_h\}, \min\{v_l, v_h\}\} = 0.5 \min\{v_l, v_h, |v_h - v_l|\}.$$

This objective is non-convex, non-concave, and not translation invariant.



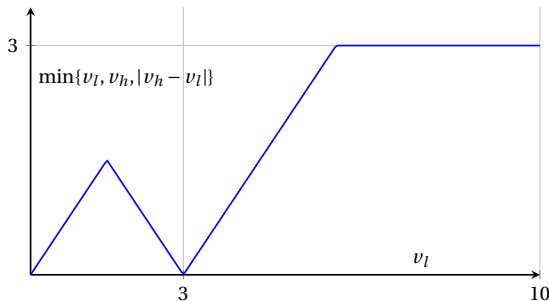
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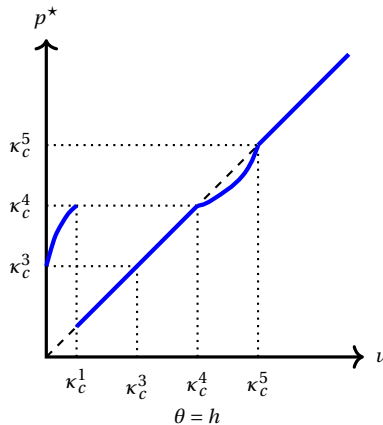
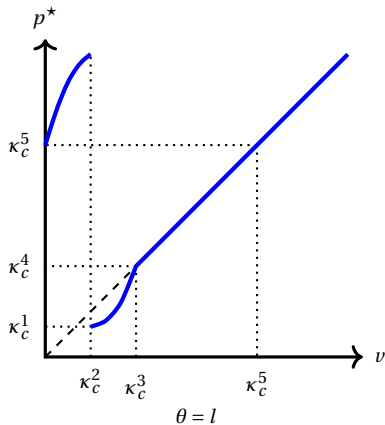
Boerma, Tsyvinski, Wang, and Zhang (2024): $|v_l - v_h|^\beta$ for $\beta \in (0, 1)$.

Moscariello (2025): $\Phi(v_l - v_h)$ with Φ S-shaped.

Economic Forces Shaping the Optimal Matching

- There are two opposing economic forces:
 - 1 If both types buy it is optimal to match consumers with similar values $|v_l - v_h|$.
 - 2 If only one types buys it is optimal to minimize the value of the buyer who does not buy $\min\{v_l, v_h\}$.

Optimal Pricing Rule



Goto closed form solution in appendix.

Theorem (Optimal Pricing)

p^ is a profit-maximizing non-discriminatory pricing rule. Moreover, every undominated profit-maximizing non-discriminatory pricing rule p induces the same average surplus $\mathbb{E}[(v - p)^+ | \theta]$ for consumer of each protected characteristic θ .*

Features of the Optimal Pricing Rule

- Low value consumers are made offers which they never accept (anecdotally this is in the data..).

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- The optimal pricing rule is not monotone in values, and does not have a cutoff structure.
- Customers with the same value will be charged different prices (evidence for random prices/approval see Blattner and Nelson 2021).
- Low and high-value customers do not benefit from the non-discrimination constraint while intermediate value customers of both types benefit.

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Fix a c , $p^\star$ corresponds to the following matching scheme $\rho_c^\star$:

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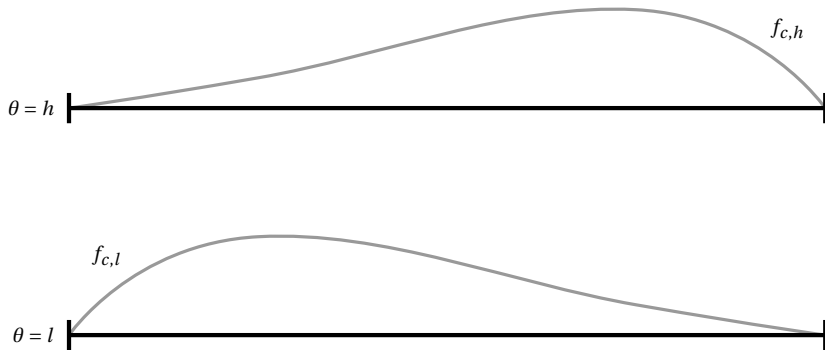
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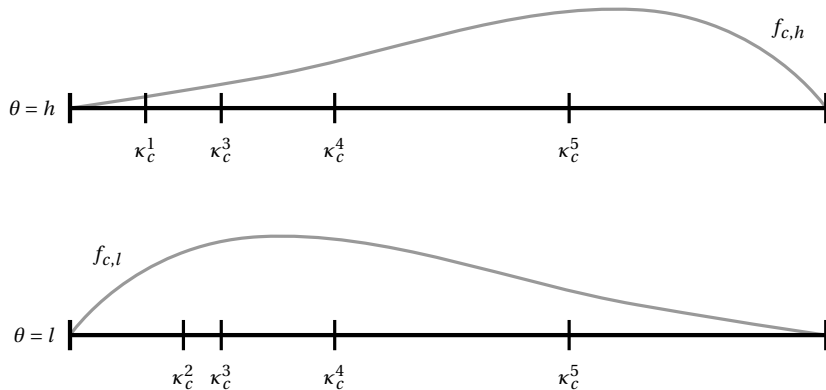
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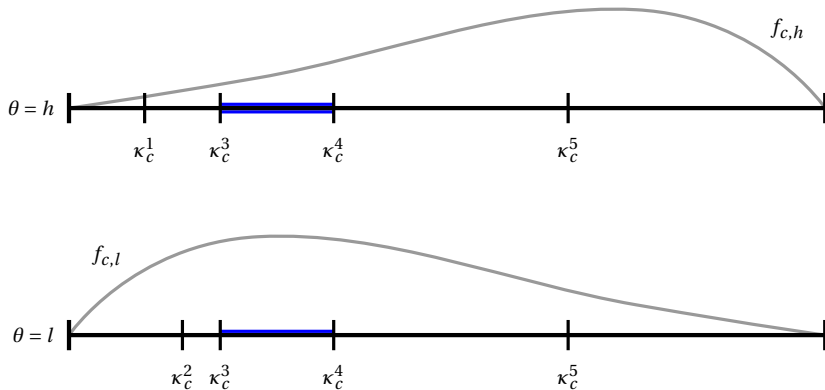
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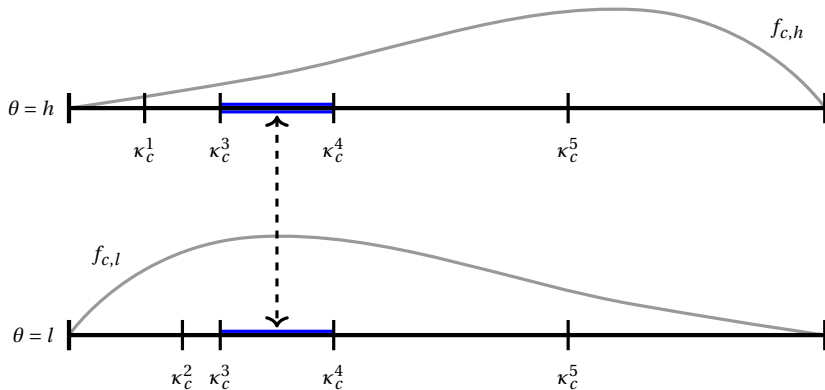
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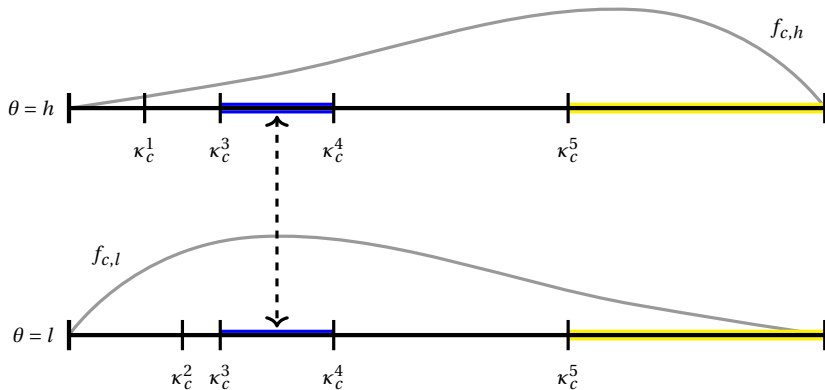
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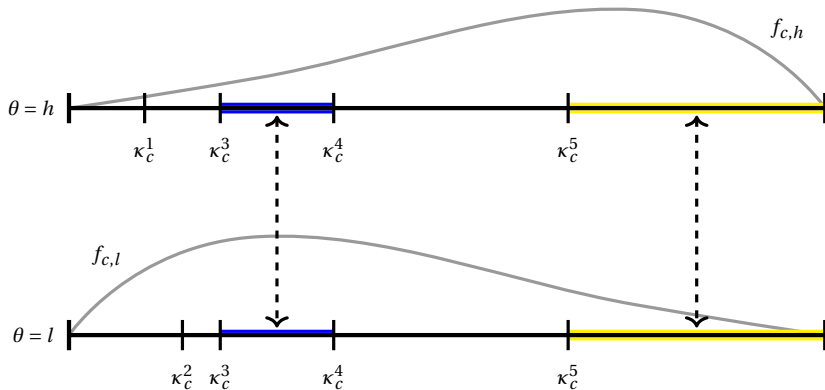
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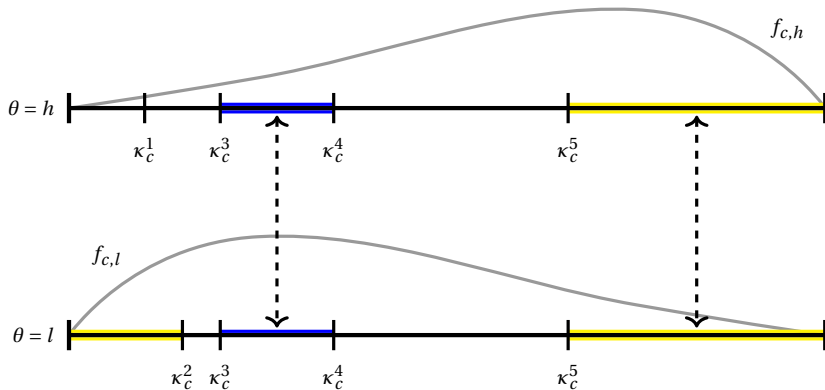
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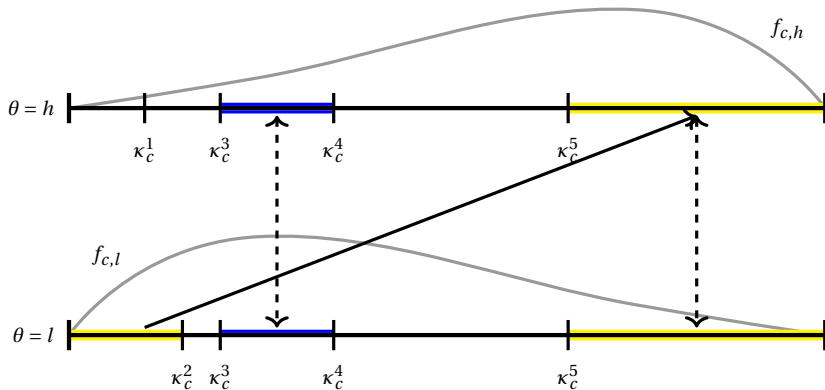
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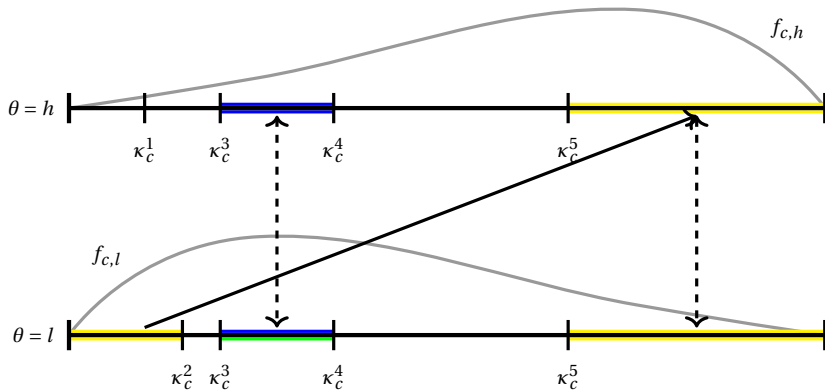
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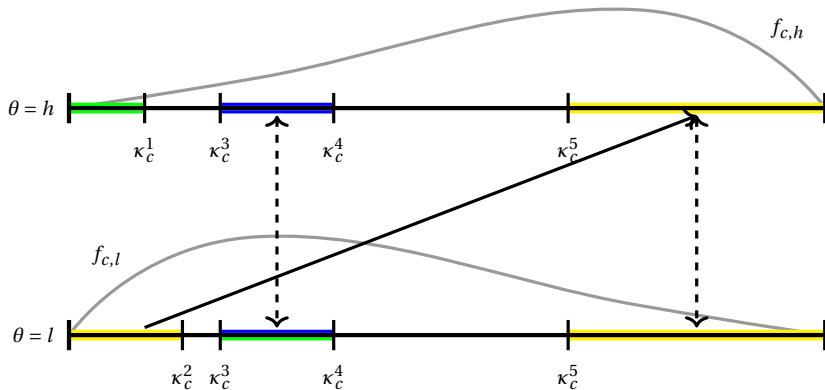
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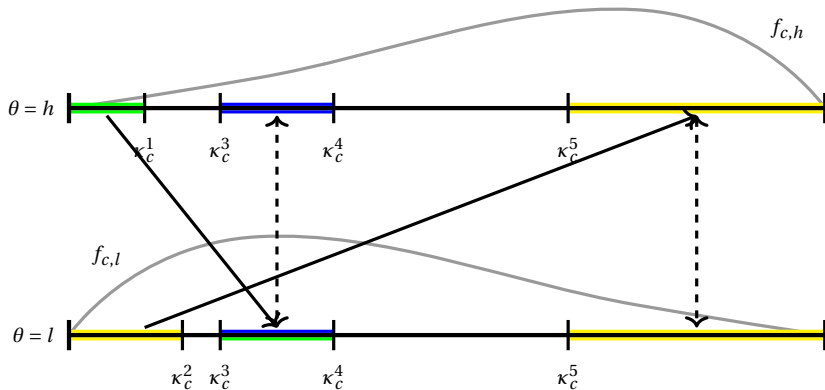
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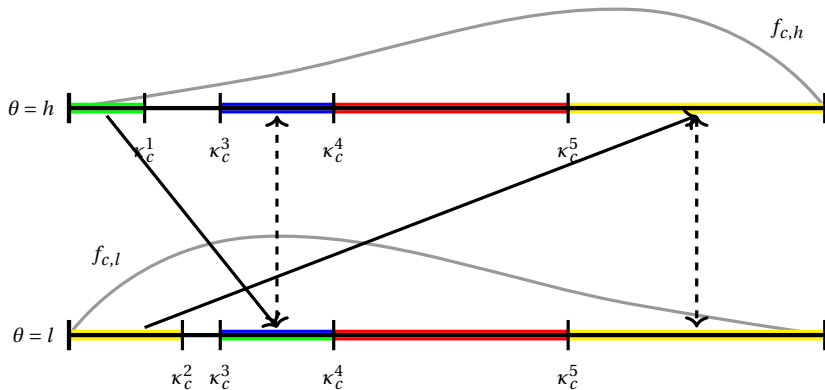
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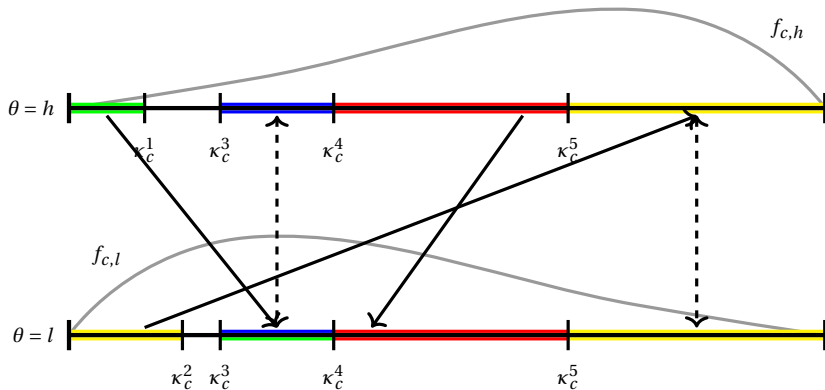
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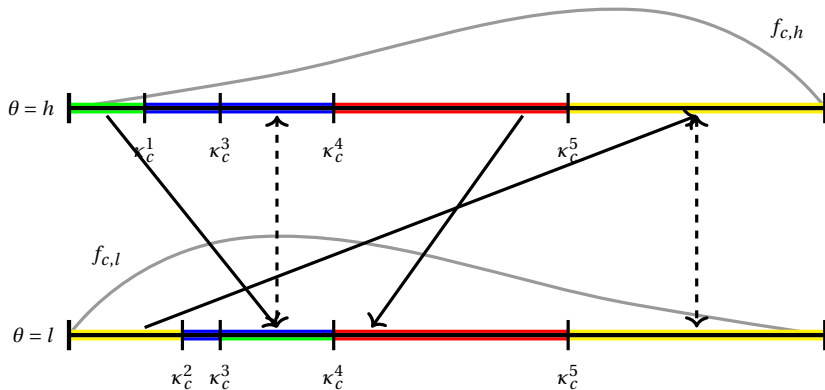
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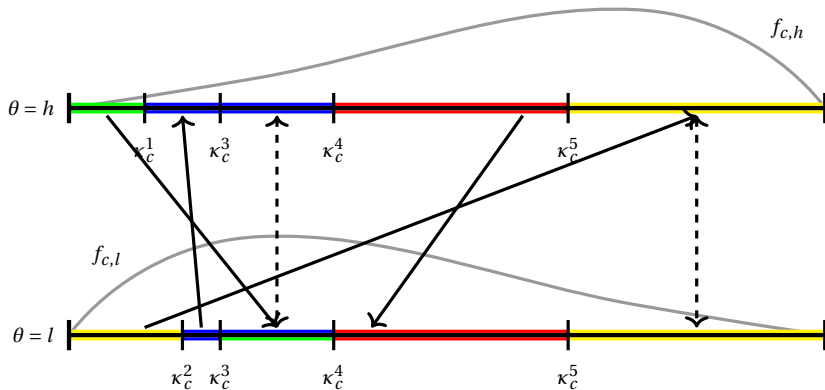
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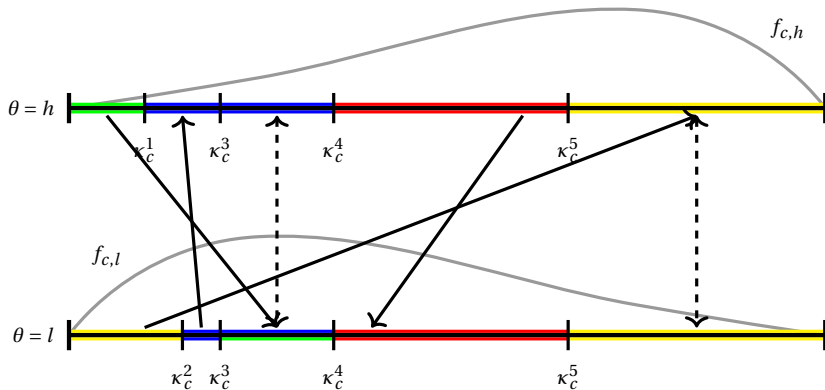
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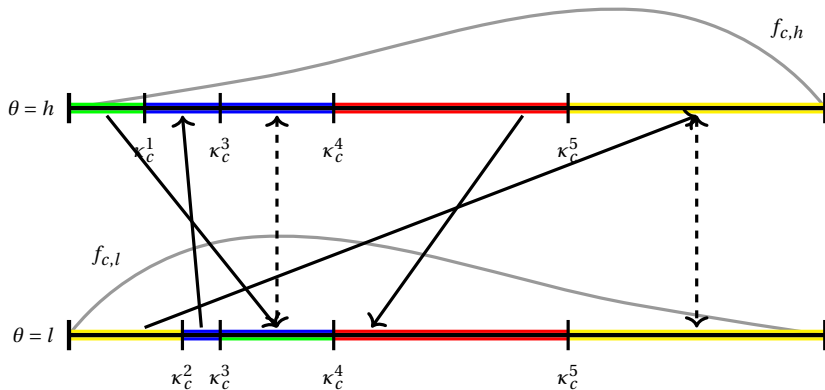
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Extreme values are matched, and charged the high price.

Welfare Implications

Proposition

Under the optimal pricing rule $p^\star$,

- 1 *Consumer surplus for both protected characteristics $\theta \in \{l, h\}$ is positive:*

$$\mathbb{E}[(v - p^\star)^+ \mid \theta = t] > 0, \forall t \in \{l, h\}.$$

- 2 *Deadweight losses from both protected characteristics are positive:*

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Benefits are differentiated:

- Low-value consumers are priced out of the market.
- High-value consumer surplus extracted.
- Intermediate value consumers enjoy surplus.

Who Benefits More from Non-Discriminatory Regulations

Proposition

For all $c \in C$, suppose that $F_{c,l}$ and $F_{c,h}$ are fixed and the share of $\theta = h$ consumers increases. Then, under the optimal pricing rule $p^\star$,

- ❶ $\theta = h$ consumer surplus decreases.
- ❷ $\theta = l$ consumer surplus increases.
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Even if there are equally many l and h types either group can benefit more.

Alternative Notions of Non-Discrimination

Non-Discriminatory Outcomes vs Non-Discriminatory Prices

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Proposition

- 1 *The unique optimal pricing rule with non-discriminatory outcomes corresponds to the assortative matching.*
- 2 *The profit of the seller is lower under equal outcomes than under non-discriminatory prices.*
- 3 *$\theta = l$ consumers are worse off under non-discriminatory outcomes than under non-discriminatory prices.*
- 4 *$\theta = h$ consumers are better off under non-discriminatory outcomes than under non-discriminatory prices.*

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- However, given the optimal pricing policy the l group will be worse off.
- One legal/conceptual difficulty with non-discriminatory outcomes is that the firm can not control the consumers decision and thus should not be punished based on them.

Extensions

- 1 Characterizing all Welfare Outcomes (appendix)
- 2 General Optimal Pricing Rule (appendix)
- 3 Commitment vs No Commitment to Prices (appendix)
- 4 Vertically Differentiated Products
- 5 Imperfect Personalized Pricing
- 6 Competition
- 7 Beyond Seller-Optimal Pricing

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- As in the baseline model, the problem reduces to optimal transport. For a matched pair (v_l, v_h) ,

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- Hence the seller solves

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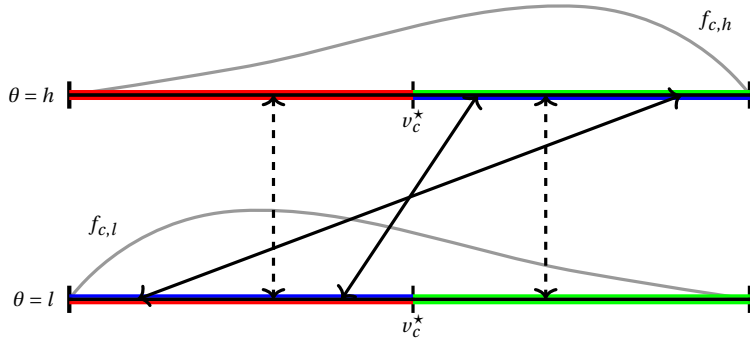


Figure: Optimal matching scheme ρ^* : The matching ρ^* matches all consumers with the same values until the mass of one group is exhausted (dashed arrows), and then match the remaining parts negatively assortatively (solid arrows).

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Interpretation:

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- Matching equal values is attractive because it allows full rent extraction.
- Among the remaining consumers, likelihood-ratio dominance leaves lower values on the l side and higher values on the h side; negative assortative matching relaxes the high-type incentive constraint and raises rents from $\theta = h$ consumers.

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Relative to the baseline linear-pricing model, nonlinear screening selects a particular matching: equal-value bunching first, then negative assortative matching for the residual mass.

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Can still solve the optimal transport and derive the optimal pricing rule.

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For example, $c = 0$, $\alpha_c = 1/2$, $w \mid v \sim \text{Unif}[0, 2v]$.

Then

$$\tilde{\pi}_c(v_l, v_h) = \max \left\{ \frac{v_l}{4}, \frac{v_h}{4}, \frac{v_l v_h}{v_l + v_h} \right\}.$$

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The solution of this problem has a similar structure as $\rho_c^\star$.

Equal Treatment Across Costs

Equal Treatment Across Costs

- So far, the price distribution could be correlated with costs.
- In some applications, prices must be independent of protected characteristics, even when the correlation only arises through costs.
- Example: unisex pricing in insurance (e.g. EU rules after *Test-Achats v. Belgium*).
- Assume v and c are perfectly correlated:

$$v - c = s \quad \text{a.s.}$$

In the insurance interpretation, s is the common risk premium consumers are willing to pay.

- Non-discrimination requires unconditional price independence:

$$\Pr[p(\omega) \leq x \mid \theta = h] = \Pr[p(\omega) \leq x \mid \theta = l].$$

- The seller therefore chooses a matching ρ between $\theta = l$ and $\theta = h$ consumers, and then sets the optimal price within each matched pair.
- For a matched pair (v_l, v_h) , profits simplify to

$$\pi(v_l, v_h) = \max \left\{ s - \frac{1}{2} |v_h - v_l|, \frac{s}{2} \right\},$$

so the seller solves

$$\max_{\rho \in \mathcal{R}} \int \pi(v_l, v_h) d\rho.$$

Optimal Matching

- Let $\underline{\kappa} \leq \bar{\kappa}$ be the unique pair such that

$$\Delta(\underline{\kappa}) = \Delta(\bar{\kappa}), \quad \underline{\kappa} + \bar{\kappa} = s.$$

- Since Δ is quasi-concave,

$$\underline{\kappa} \leq v^* \leq \bar{\kappa}, \quad f_h(v) \geq f_l(v) \text{ for all } v \geq \bar{\kappa}.$$

- The optimal transition rule $\gamma^*(\cdot | v_h)$ has three regions:

- $v_h \leq \underline{\kappa}$: assortative matching, $v_l = v_h$;
- $v_h \in (\underline{\kappa}, \bar{\kappa}]$: match to

$$v_l = F_l^{-1}(F_h(v_h) + \Delta(\underline{\kappa}));$$

- $v_h > \bar{\kappa}$: randomize between $v_l = v_h$ and

$$v_l = F_l^{-1}(\Delta(\bar{\kappa}) - \Delta(v_h)),$$

with weights chosen so that the marginal on v_l is F_l .

- Define

$$\rho^*(A, B) = \int_B \gamma^*(A | v_h) F_h(dv_h).$$

Then $\rho^* \in \mathcal{R}$ and has marginals F_l and F_h .

- Proposition:** ρ^* solves

$$\max_{\rho \in \mathcal{R}} \int \pi(v_l, v_h) d\rho.$$

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- Since every consumer already faces the competitive price, protected characteristics do not affect markups or rents.
- In that benchmark, there is no role for non-discrimination regulation through pricing restrictions.
- The need for regulation thus arises from a lack of competition, if firms behavior is revenue maximzing.

Beyond Seller Optimal Segmentations

So far focused on seller optimal segmentations.

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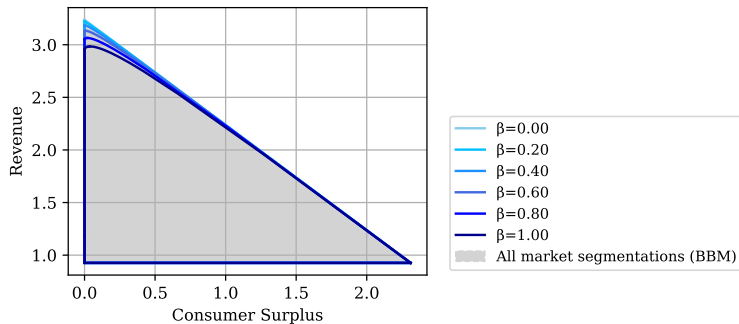
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Analytic solution for 2 values.

Feasible Welfare Outcomes



$$c = 0, \alpha_c = 1/4,$$

$$F_{c,h} = (\beta + 1/4(1 - \beta)) \times \exp(10) + (1 - (\beta + 1/4(1 - \beta))) \times \exp(1)$$

$$F_{c,l} = 1/4(1 - \beta) \times \exp(10) + (1 - 1/4(1 - \beta)) \times \exp(1)$$

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Either protected characteristics could benefit more. High cost consumers don't benefit.

Thank you!

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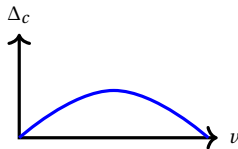
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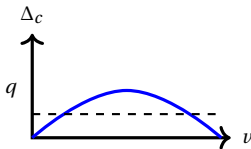


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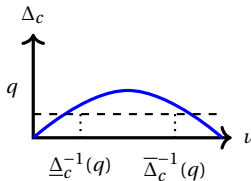


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Lemma

For any c , there exists a unique increasing vector $\kappa_c \in \mathbb{R}_+^5$

$$\kappa_c^2 = F_{c,l}^{-1}(\Delta_c(\kappa_c^3) + F_{c,h}(\kappa_c^1)) = F_{c,l}^{-1}(\Delta_c(\kappa_c^4)) = F_{c,l}^{-1}(\Delta_c(\kappa_c^5))$$

$$\kappa_c^1 - c = (1 - \alpha_c) \cdot (\kappa_c^3 - c) = \alpha_c \cdot (\kappa_c^5 - \kappa_c^4).$$

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Let $p^\star$ be defined as

$$p^\star(v, c, l, r) := \begin{cases} \overline{\Delta}_c^{-1}(\Delta_c(\kappa_c^5) - F_{c,l}(v)), & \text{if } v < \kappa_c^2 \\ F_{c,h}^{-1}(F_{c,l}(v) - F_{c,l}(\kappa_c^2) + F_{c,h}(\kappa_c^1)), & \text{if } v \in [\kappa_c^2, \kappa_c^3) ; \\ v, & \text{if } v \geq \kappa_c^3 \end{cases}$$

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Sketch of Proof

Optimal Segmentation as Optimal Transport

$\mathcal{R}_c$: set of joint distributions $\rho_c \in \Delta(V^2)$ with marginals $F_{c,l}$ and $F_{c,h}$.

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Optimal Transport Problem:

$$\max_{\rho_c \in \mathcal{R}_c} \int_{V^2} \pi_c(v_l, v_h) d\rho_c.$$

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Dual problem:

$$\pi_{\star}(c) = \inf_{\phi, \psi} \left[\int_0^{\infty} \phi(v_l) F_{c,l}(dv_l) + \int_0^{\infty} \psi(v_h) F_{c,h}(dv_h) \right] \quad (\text{D})$$

subject to

$$\phi(v_l) + \psi(v_h) \geq \pi_c(v_l, v_h)$$

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Lemma (Kantorovich Duality)

$\pi_(c) = \pi^*(c)$ for all c . Moreover, for any feasible ρ in (P) and for any feasible ϕ_c, ψ_c in (D), ρ_c is a solution of (P) and (ϕ, ψ) is a solution of (D) if and only if $\phi_c(v_l) + \psi_c(v_h) = \pi_c(v_l, v_h)$ for all $(v_l, v_h) \in \text{supp}(\rho_c)$.*

Solution to the Dual Problem

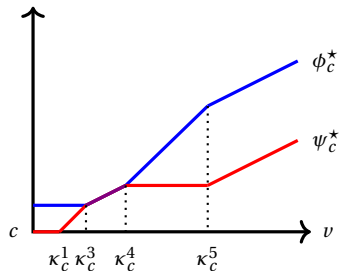
$$\phi_c^*(v_l) := \begin{cases} \kappa_c^1 - c, & v_l \leq \kappa_c^3 \\ (1 - \alpha_c)(v_l - c), & v_l \in (\kappa_c^3, \kappa_c^4] \\ v_l - c - \alpha_c(\kappa_c^4 - c), & v_l \in (\kappa_c^4, \kappa_c^5] \\ (1 - \alpha_c)(v_l - c) + \kappa_c^1 - c, & v_l > \kappa_c^5 \end{cases};$$

$$\psi_c^*(v_h) := \begin{cases} 0, & v_h \leq \kappa_c^1 \\ v_h - \kappa_c^1, & v_h \in (\kappa_c^1, \kappa_c^3] \\ \alpha_c(v_h - c), & v_h \in (\kappa_c^3, \kappa_c^4] \\ \alpha_c(\kappa_c^4 - c), & v_h \in (\kappa_c^4, \kappa_c^5] \\ \alpha_c(v_h - c) - (\kappa_c^1 - c), & v_h > \kappa_c^5 \end{cases}.$$

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For all c and for all (v_l, v_h) ,

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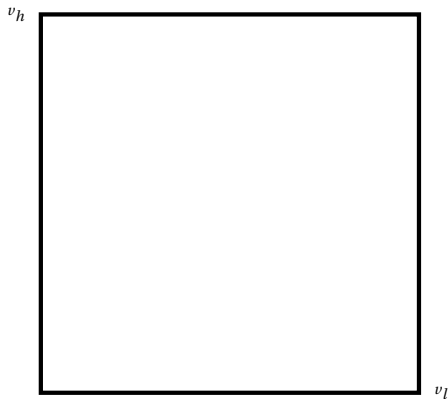
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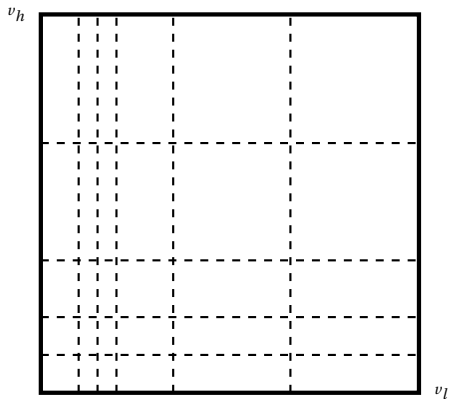
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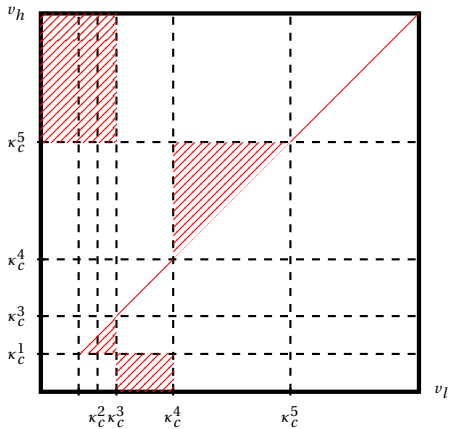
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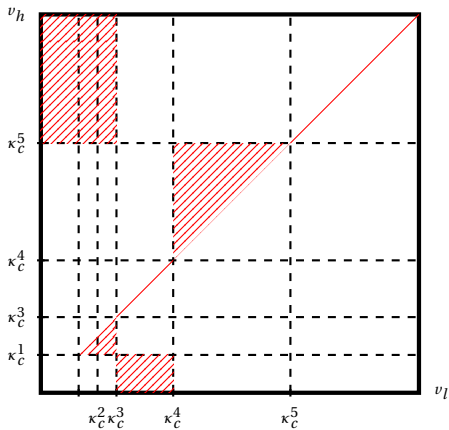
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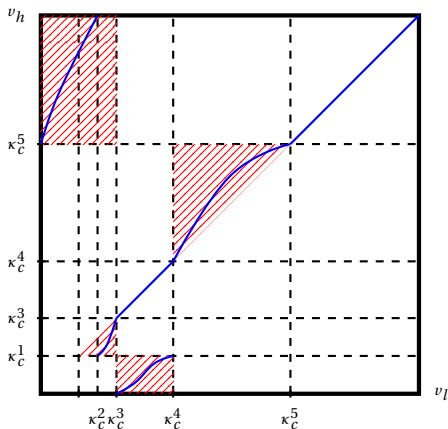
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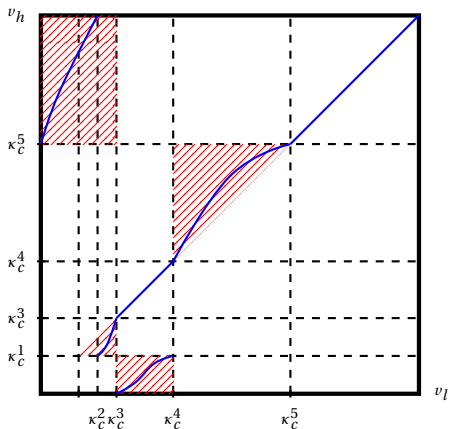


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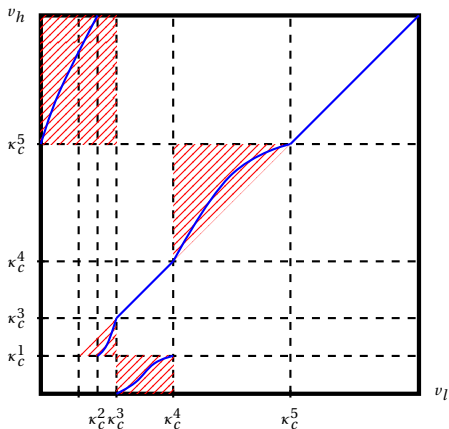


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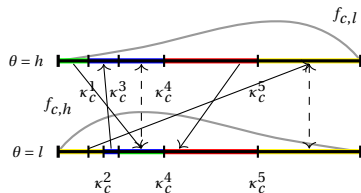
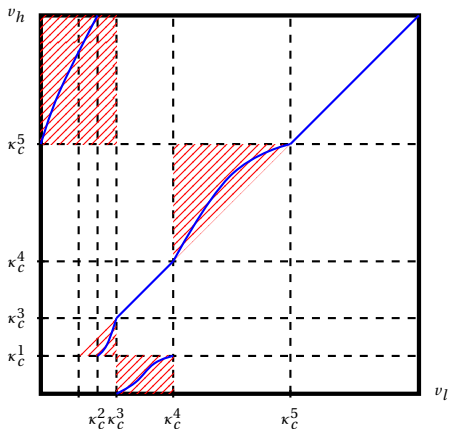


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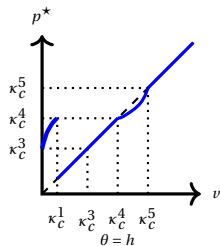
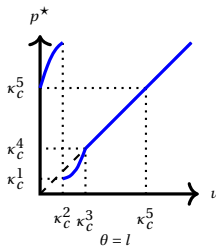
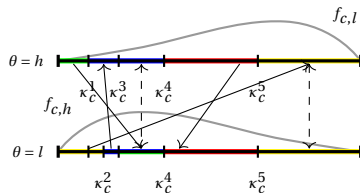
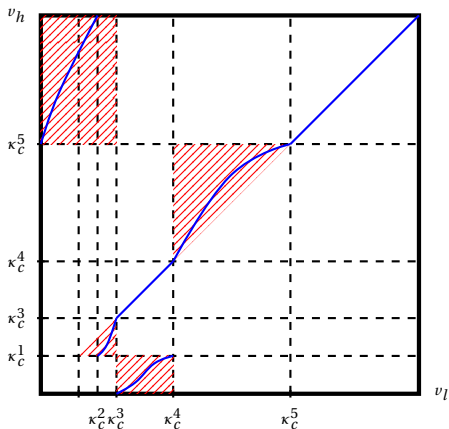


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Theorem (Welfare Outcomes)

For all $c \in C$, $(\sigma_l, \sigma_h) \in \Sigma_c^$ if and only if*

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Welfare outcomes under optimal non-discriminatory pricing rules are Pareto ranked, where $\theta = l$ consumers retain zero surplus in the worse case.

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Can extract all gains from trade when Assumption fails.

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C_1, C_2, C_3 is a partition of C .

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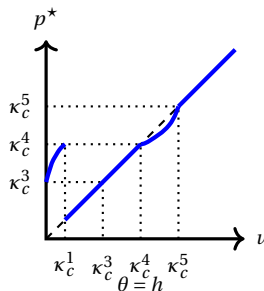
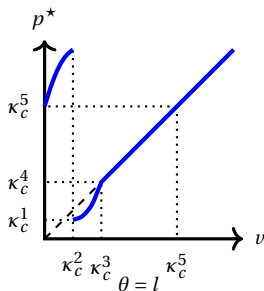
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General Optimal Pricing Rule

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If $c \in C_2$, let $\eta_c^l \leq c$ be such that $F_{c,l}(c) - F_{c,l}(\eta_c^l) = \Delta_c(v_c^\star)$; $\eta_c^h := F_{c,h}^{-1}(F_{c,l}(\eta_c^l))$.

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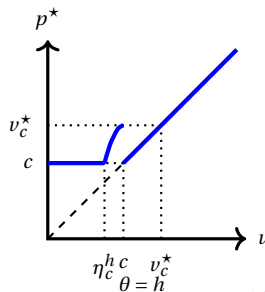
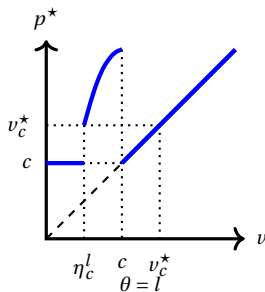
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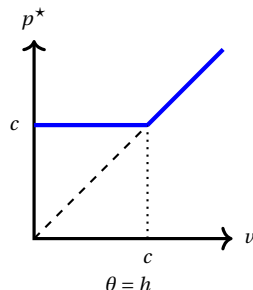
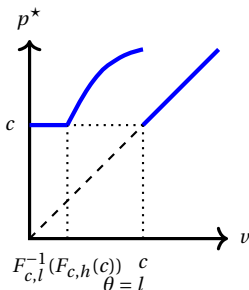
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General Optimal Pricing Rule

Theorem

p^* is a profit-maximizing non-discriminatory pricing rule. Moreover, every undominated profit-maximizing non-discriminatory pricing rule p induces the same average surplus $\mathbb{E}[(v - p)^+ | \theta]$ for consumers of each protected characteristic θ .

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Given a segmentation s , seller's profit is

$$\tilde{\Pi}(s) := \mathbb{E}[\max_{x \geq 0} \mathbb{E}[(x - c) \mathbf{1}\{x \geq c\} \mid s]].$$

Proposition

The seller's profit under an optimal non-discriminatory segmentation equals the profit under an optimal non-discriminatory pricing rule. That is,

$$\max_{s \in \mathcal{S}} \tilde{\Pi}(s) = \max_{p \in \mathcal{D}} \Pi(p).$$

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For any non-discriminatory segmentation s , let $p := s$. Any pricing rule optimal with respect p to s must give $\Pi(p) \geq \tilde{\Pi}(s)$.

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Any such pricing rule is measurable with respect to s and thus is non-discriminatory.